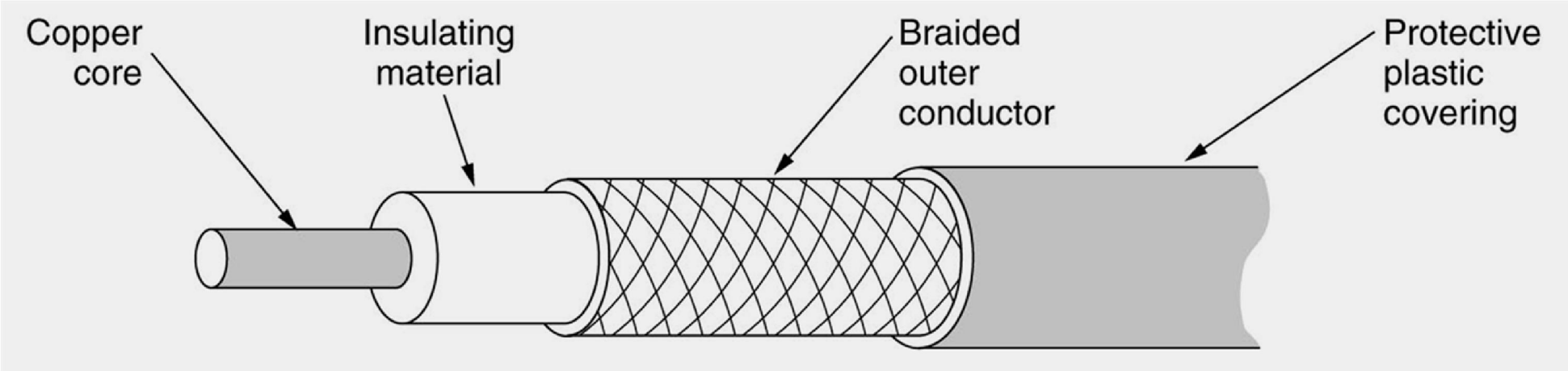
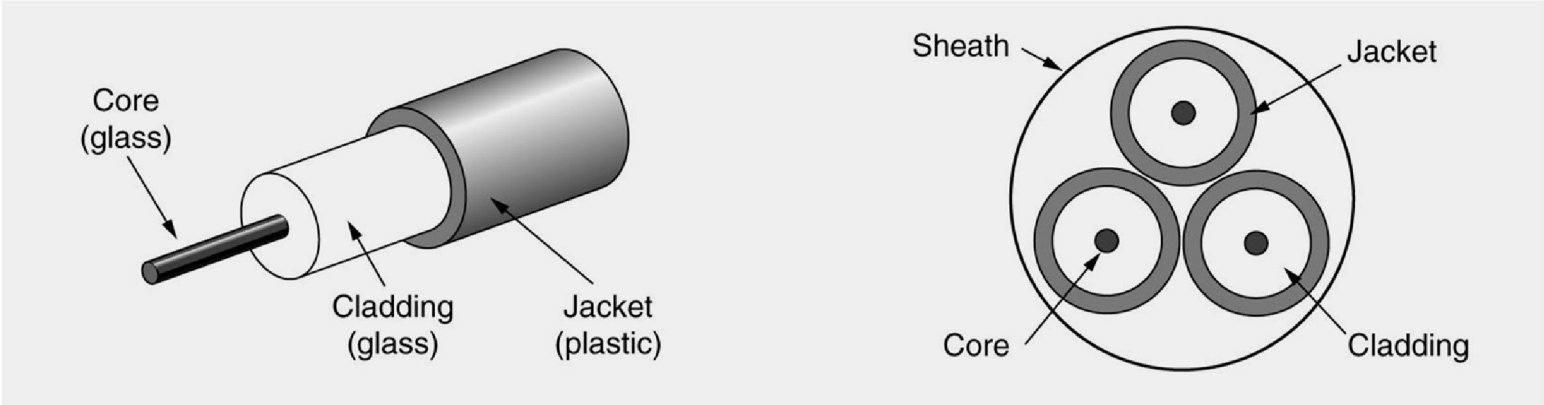
Transmission Media

- The ***Transmission Medium*** is the path along which the signal travels.
- The type of medium determines the type of signal to be used and can affect the reliability of communications.
- Conventional networks use copper wire:
  - Low resistance to electrical current,
  - The terms ***copper*** and ***wire*** are used interchangeably,
  - Prone to interference from other wires close-by,
  - This interference can be minimised by using ***Twisted Pair*** wiring and ***Co-axial*** cable.

TP, Co-ax and Optical Fibre Cables

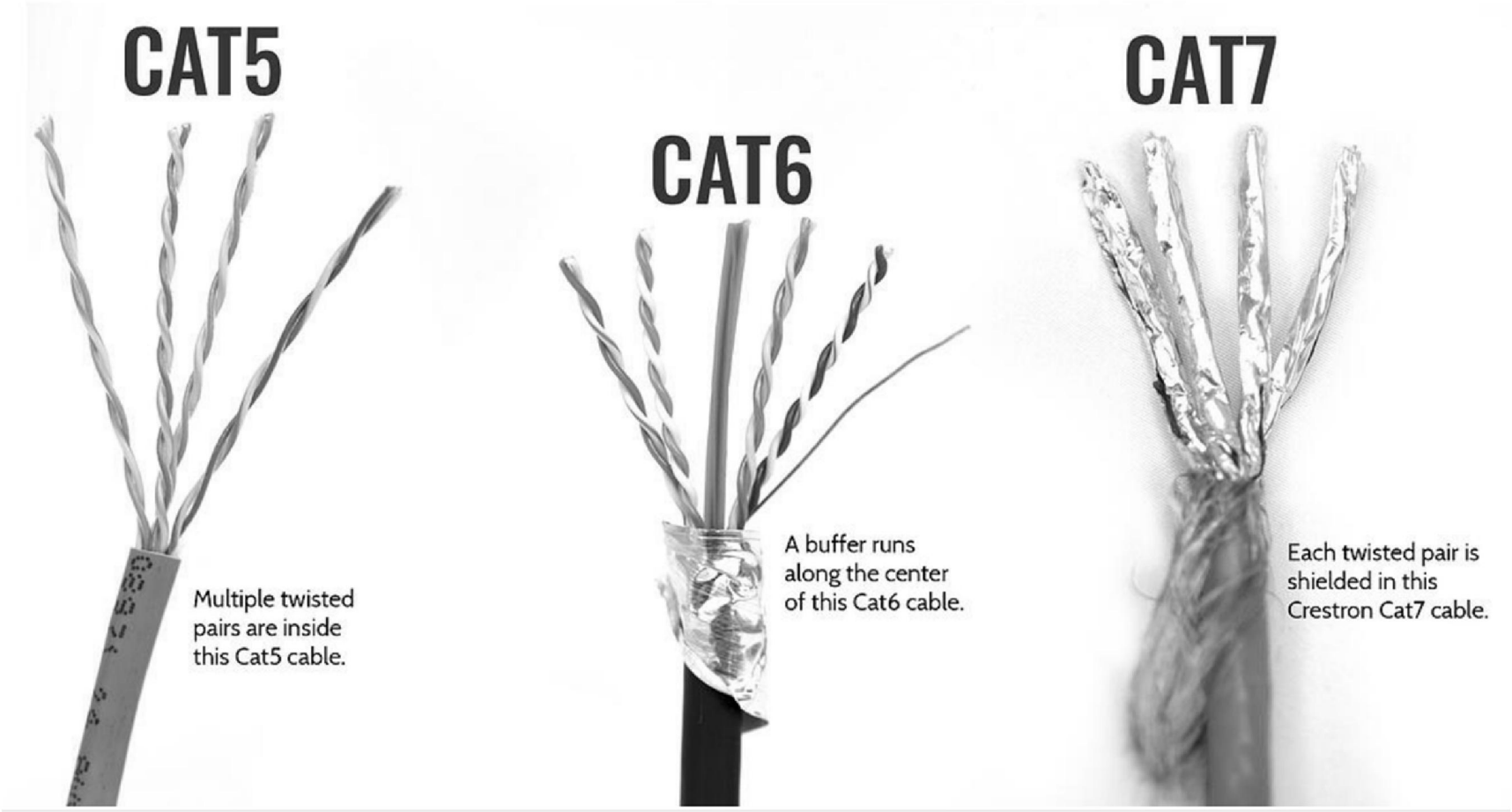




Twisted Pair

- Characteristics:
  - Comprised 2 insulated copper wires twisted together,
  - These twists reduce ***crosstalk***,
  - Typically many pairs are bundled in a protective sheath,
  - Very cheap and easy to use,
  - BW: Cat 3 = 16MHz., Cat 5 = 100MHz., Cat 6 = 250MHz.
- Applications:
  - In office buildings for phone and computer connectivity
  - Residential telephone – the local loop.

Twisted Pair Cables



Coaxial Cable

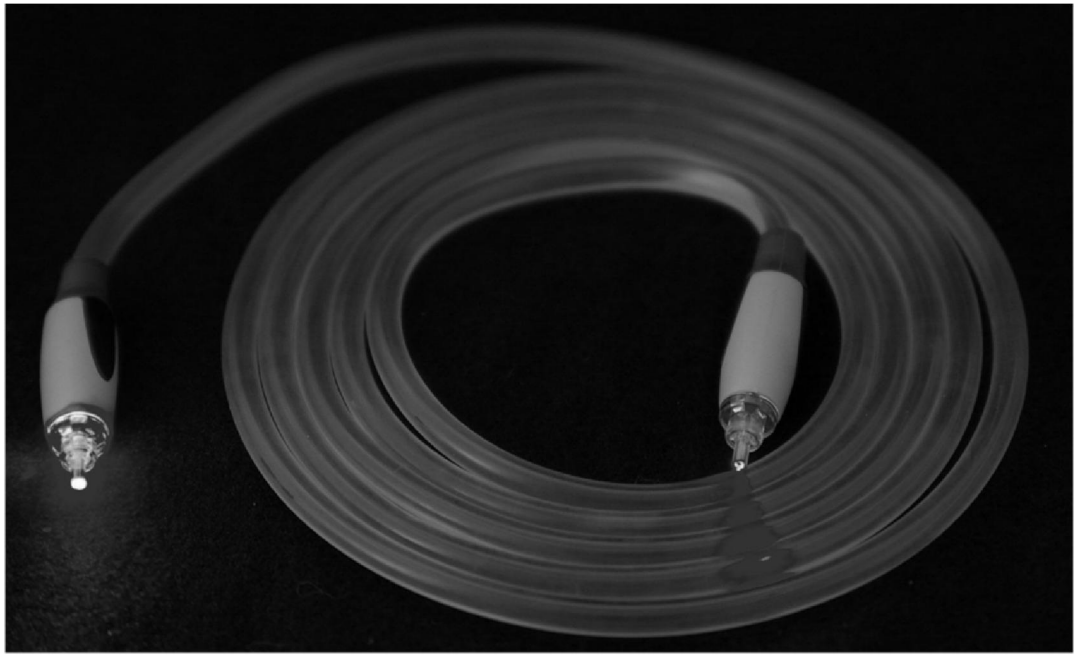
- Characteristics:
  - Comprised of central copper conductor and a hollow outer cylindrical conductor,
  - A surrounding sheath holds the inner conductors in place,
  - BW: 1GHz.
- Applications:
  - Cable TV,
  - Long distance telephone network,
  - Computer connectivity (LANs).

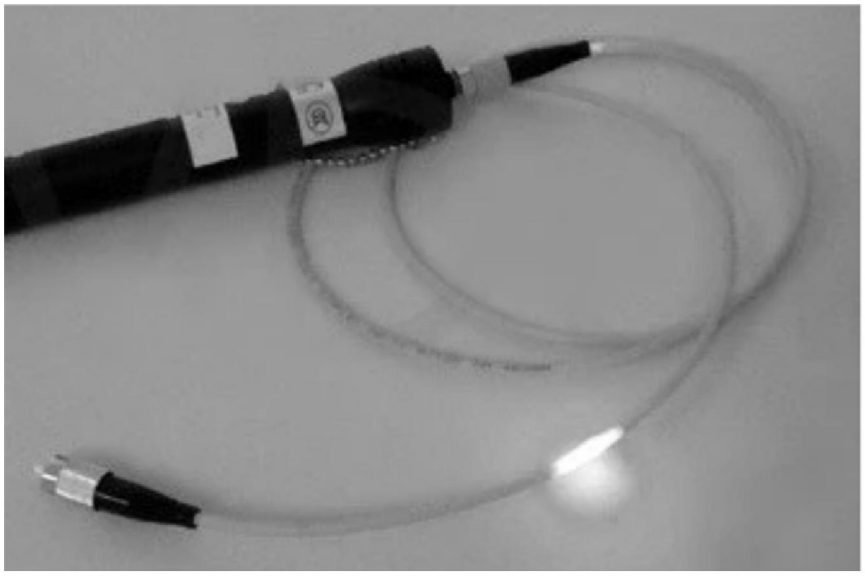
Co-axial cable

Optical Fibre

- Characteristics:
  - Comprised of thin flexible transparent glass fibre core,
  - This core is surrounded by glass or plastic cladding with a lower refractive index,
  - Outer sheath provides rigidity and protection,
  - Several fibres may be bound in a single jacket,
  - Two types of light source used:
    - Light emitting diode (LED) - low cost,
    - Injection laser diode (ILD) for higher data rates and greater distances.

Fibre Optic cable and a faulty cable





Optical Fibre

- Applications:
  - Long distance telephone connections,
  - Metropolitan trunks,
  - Rural exchange trunks,
  - Local loop (becoming more popular),
  - LANs.

Optical Fibre V Coaxial Cable

- Optical Fibre cables offer the following advantages over Co-axial cables:
  - Electromagnetic isolation i.e. no interference,
  - Lower Attenuation hence greater repeater spacing (50km v 5km for copper),
  - Greater Capacity – up to 10 Gbps over many km's,
  - Requires only single core for complete circuit:
    - Hence smaller in size, lighter in weight.

Wireless Transmission

- Two configurations:
  - *Omni-directional* e.g. radio station,
  - *Directional* e.g. microwave/satellite links.
- Radio Transmission characteristics:
  - Networks that use EM radio waves are known as ***Radio Frequency (RF)*** networks,
  - There is no physical connection between stations,
  - Transmission is achieved using an antenna.

Terrestrial Microwave

- Characteristics:
  - Microwaves travel at higher frequency than RF. Typically 2 – 40 GHz.,
  - This higher frequency allows for a larger bandwidth which implies a high data rate e.g. 120 Mbps,
  - Waves are focused into a narrow beam by a parabolic dish antenna,
  - Waves are received by a similar dish antenna,
  - Line of sight required (although not always possible),
  - Prone to interference from rainfall and other dishes,
  - Prone to misalignment in storm.

Terrestrial Microwave Links

Terrestrial Microwave

- Applications:
  - Back-to-back installations used on long haul links (this is their main application).
  - Also used where there are difficulties laying cables e.g. mountain top installations.
  - Very common with mobile phone networks e.g. the ***3 Network*** relies heavily on MW links.

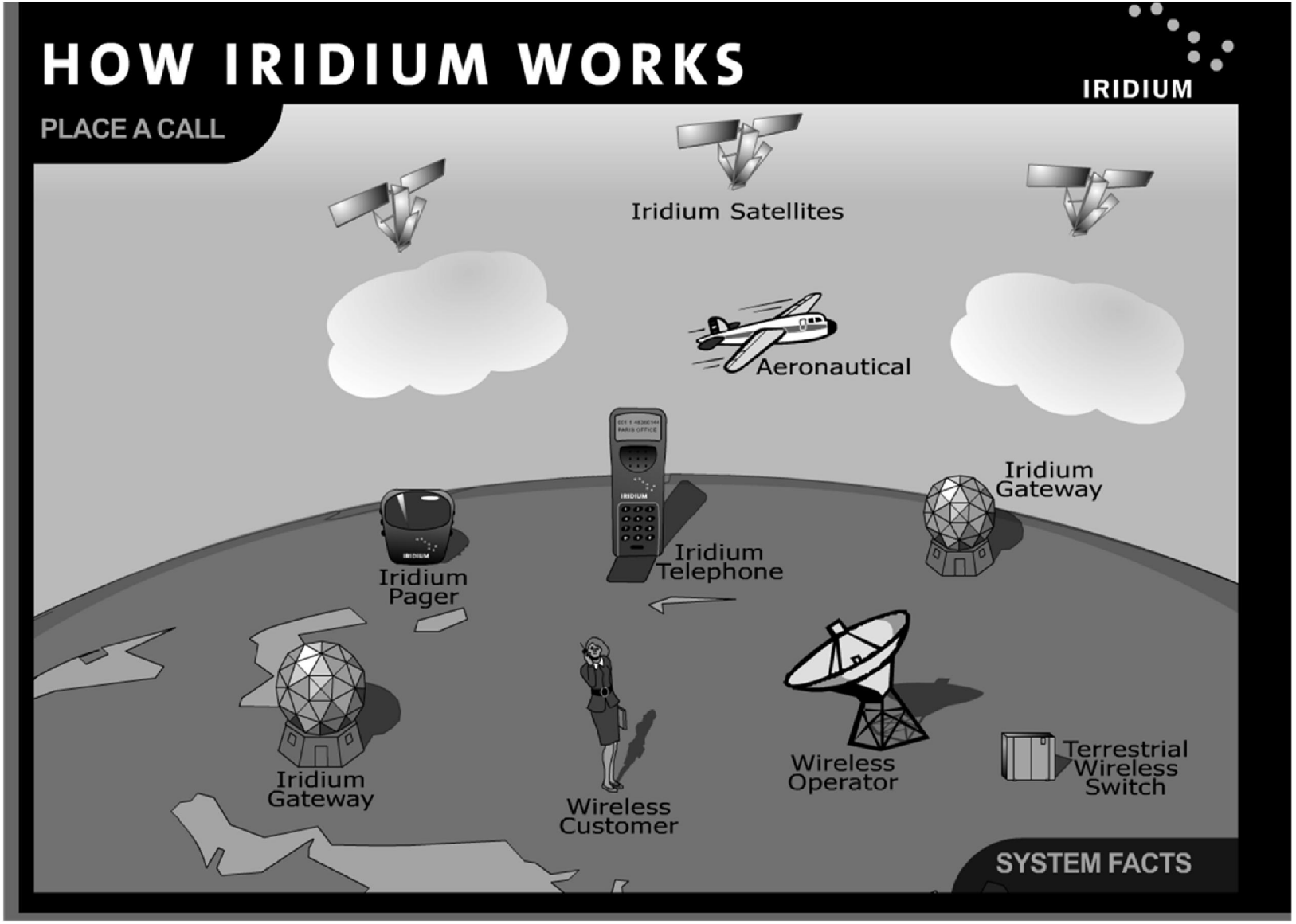
Satellite Microwave (aka Satellites)

- Satellites can link two or more ground stations – i.e. they act as a *relay,*
- Satellites use two different bands (frequency ranges) to communicate:
  - Different frequencies allow full duplex comms.,
  - There are *up* and *down* links each using different frequencies.
- Two types of Satellite (GEOs and LEOs):
  - GEOs remain stationary over a point on earth,
  - LEOs move rapidly across the sky.

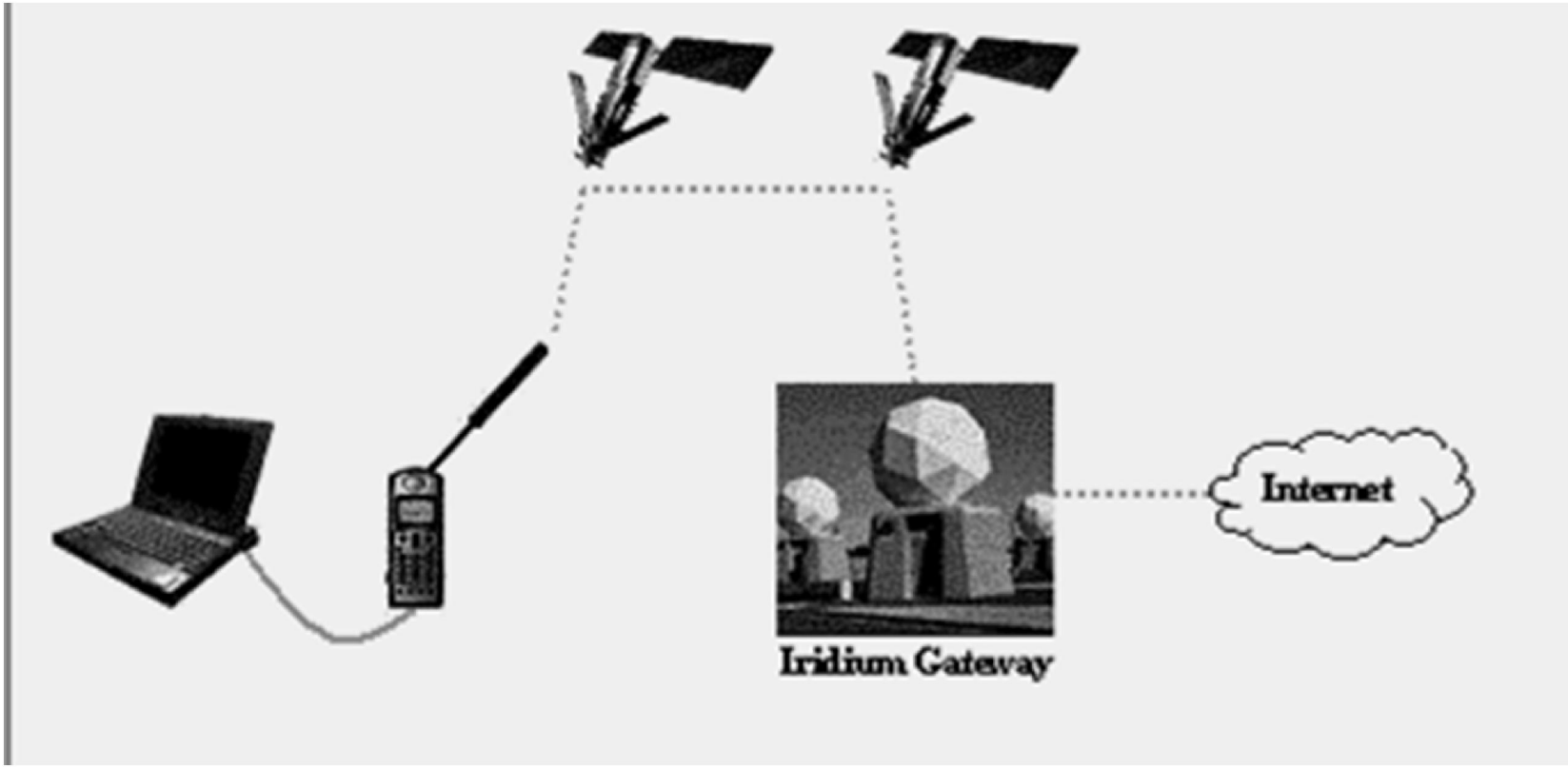
Satellites

- Characteristics:
  - Originally operated in range 1 – 10GHz.,
  - The 4/6 GHz band is now saturated,
  - 12/14 GHz now open for service.
  - Inherent propagation delay (time delay) approx. 0.25sec.
- Applications:
  - Television distribution,
  - Long-distance telephone transmission,
  - Private business networks.

The Iridium Satellite Network



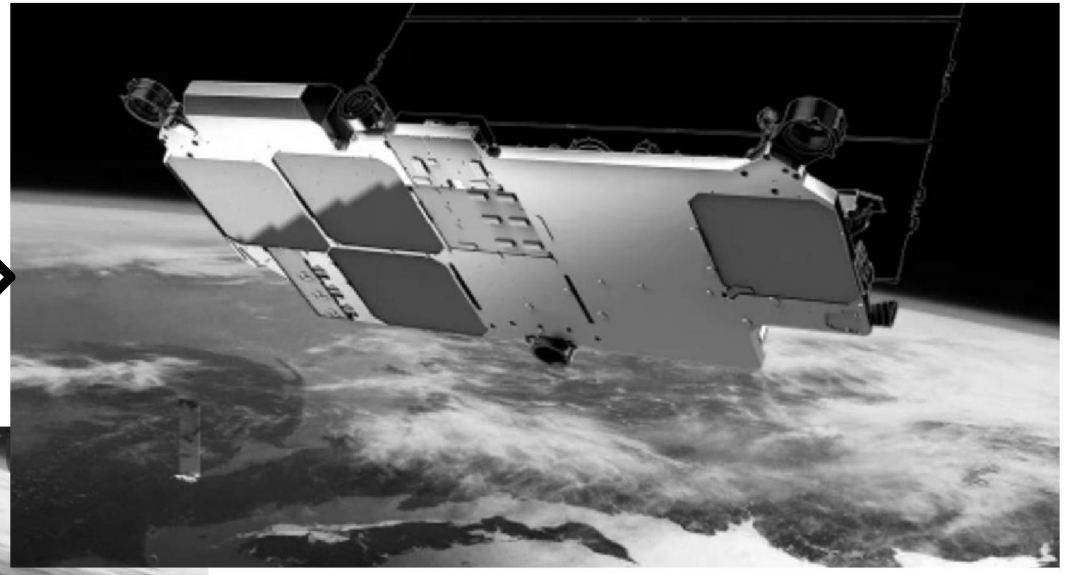
Using Iridium Network for Computer Comms.

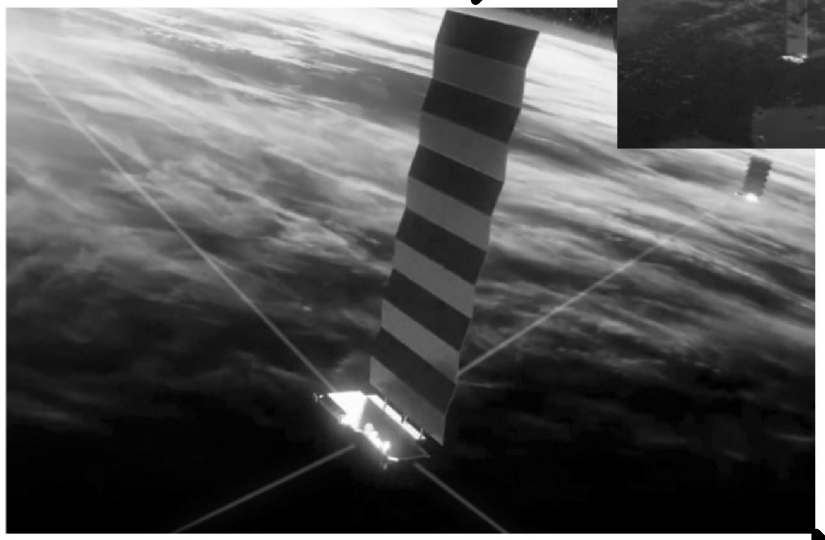


Elon Musk's Starlink Network

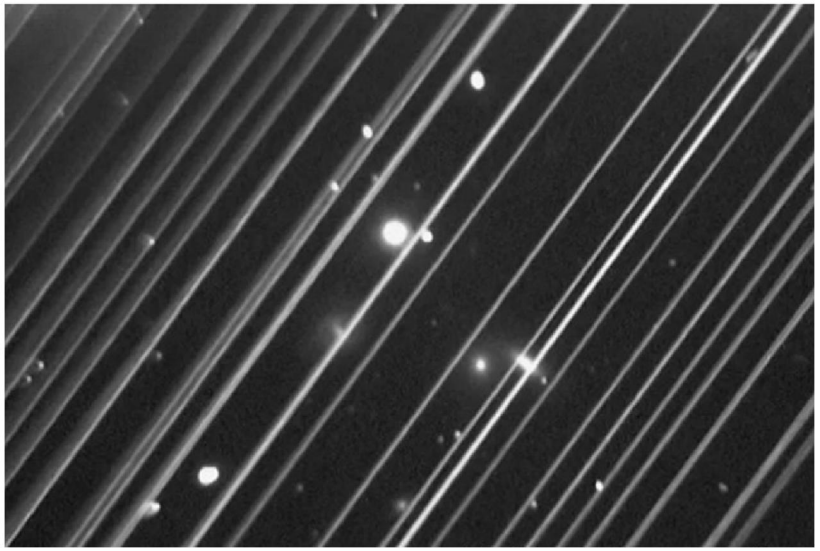
Falcon 9
Launcher

Starlink
Satellite





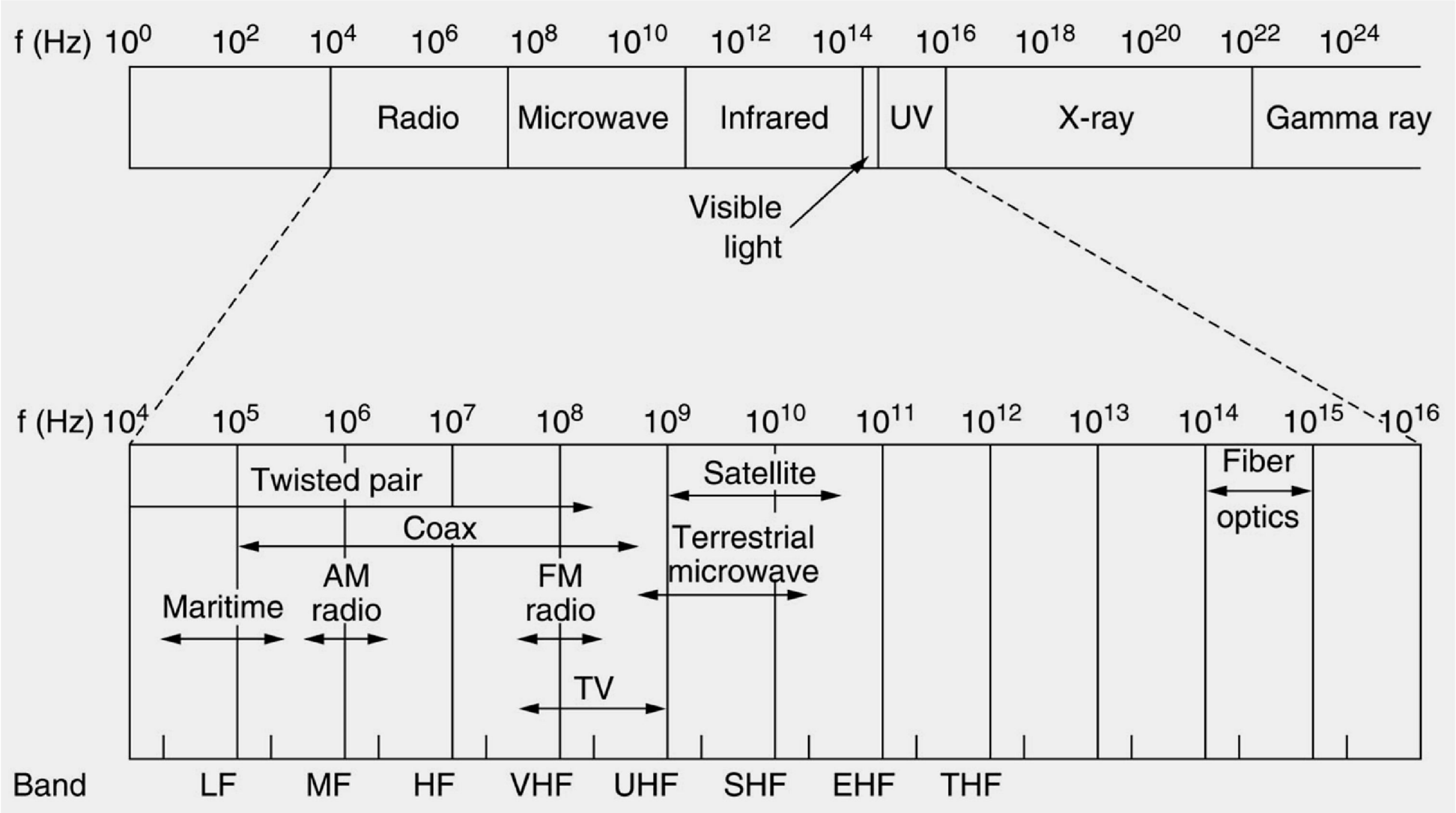
Environmental Impact



Infrared

- Characteristics:
  - Does not require antennae or special license to operate,
  - Uses LED,
  - Very low cost,
  - Cannot penetrate obstacles i.e. requires line of sight.
- Applications:
  - Remote controls,
  - Computer-to-computer connectivity e.g mobile phones etc.

Bluetooth

- Characteristics:
  - Uses the 2.4 GHz. ISM band (Industrial, Scientific and Medical),
  - Uses low power implies short range (approx. 10m),
  - Comprises 79 channels of 1MHz. each,
  - Uses a type of encoding called FSK which provides for 1 bit per Hz. i.e. 1Mbps per channel,
- Applications:
  - Connection of a variety of computing devices

The E-M Spectrum